General Chemistry (Chem1006)

Chapter 5

Electronic Structure & Periodic Properties of Elements

5.1. Electromagnetic energy

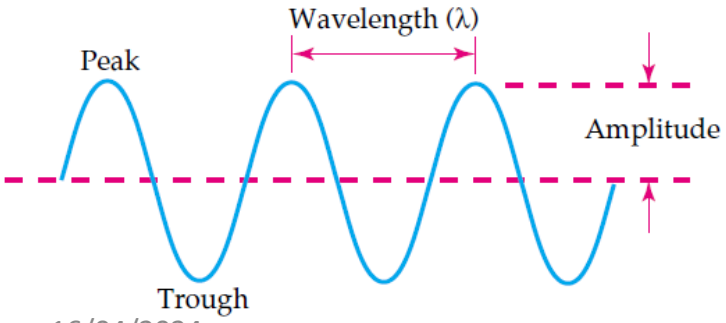
- The are many types of electromagnetic radiation in addition to visible light.
  - The radio waves that carry music to our radios and mobile phone,
  - The energy used to cook food in your microwave,
  - The infrared radiation (Heat) from a glowing fireplace,
  - X-rays used by a dentist and

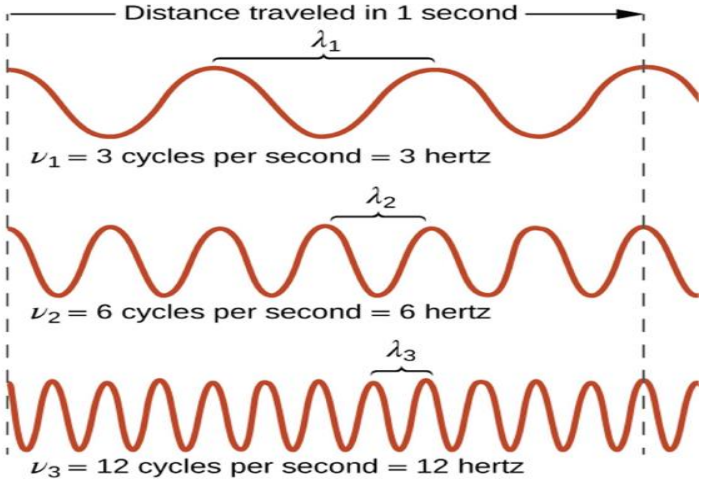
- All are forms of electromagnetic radiation share certain fundamental characteristics.

5.1.1. Characteristics of Light

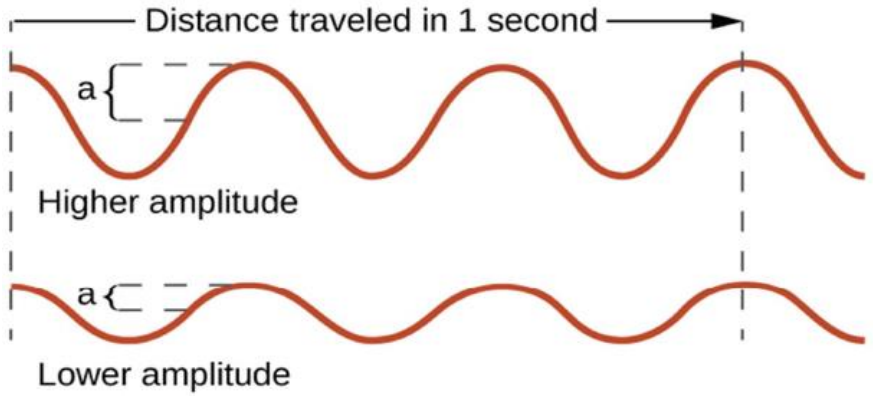
Waves

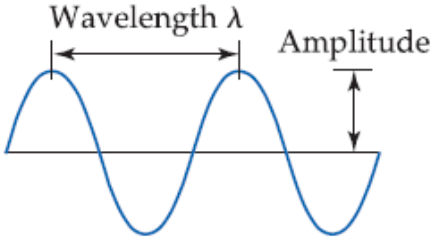
- All types of electromagnetic radiation, can be described in the terminology of waves.
- To help characterize any wave, we specify its **wavelength** (denoted by λ, the lowercase Greek letter lambda), a **frequency** (denoted by ν, the lowercase Greek letter nu), and an **amplitude.**
- **wavelength** is the distance between two consecutive peaks or troughs in a wave (measured in meters in the SI system).
- It ranges from 10^3 m to 10^{-12} m. **frequency** is the number of wave cycles that pass a specified point in space in a specified amount of time (in the SI system, this is measured in seconds).
  - ✓ A cycle corresponds to one complete wavelength.
  - ✓ The unit for frequency, expressed as cycles per second [s^{-1}], is the hertz (Hz) Common multiples of this unit are megahertz, (1 MHz = 1 × 10^6 Hz) and gigahertz (1 GHz = 1 × 10^9 Hz).



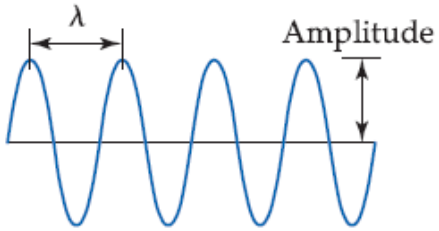


- **amplitude** corresponds to one-half the height between the peaks and troughs.
  - ✓ It is related to the intensity of the wave, which for light is the brightness, and for sound is the loudness.

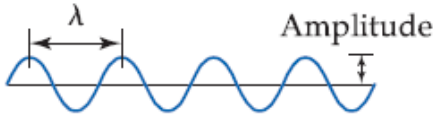




(a) Two complete cycles of wavelength λ



(b) Wavelength half of that in (a); frequency twice as great as in (a)



(c) Same frequency as (b), smaller amplitude

- **Multiplying** the wavelength of a wave in meters (m) by its frequency gives the speed of the wave in meters per second (m/s).

- The rate of travel of all electromagnetic radiation in a vacuum is a constant value, commonly called the speed of light and abbreviated **c**.
  - ✓ It's numerical value is defined as exactly usually rounded off to **3×10^8 m/s.**

$$\text{Wavelength} \times \text{Frequency} = \text{Speed}$$

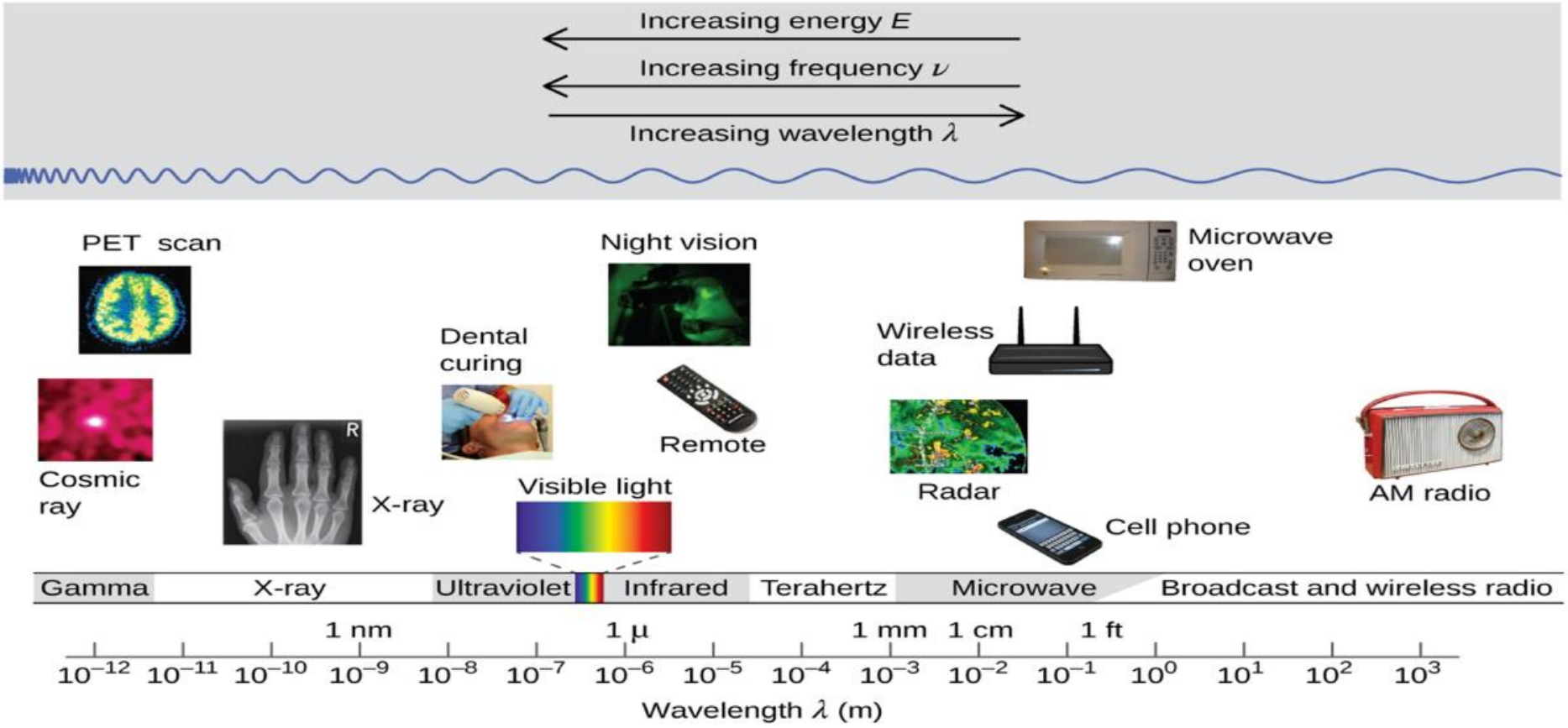
$$\lambda\,(\text{m}) \times \nu\,(\text{s}^{-1}) = c\,(\text{m/s})$$

which can be rewritten as:

$$\lambda = \frac{c}{\nu} \qquad \text{or} \qquad \nu = \frac{c}{\lambda}$$

- frequency and wavelength are **inversely related**

- This equation says that frequency and wavelength are **inversely related**:
  - ✓ Electromagnetic radiation with a long wavelength has a low frequency, and radiation with a short wavelength has a high frequency.
  - ✓ For example, radio waves are usually specified as frequencies (typically in units of MHz), while the visible region is usually specified in wavelengths (typically in units of nm or angstroms).



Determining the Frequency and Wavelength of Radiation

A sodium streetlight gives off yellow light that has a wavelength of 589 nm (1 nm = 1×10^{-9} m). What is the frequency of this light?



Solution

We can rearrange the equation $c = \lambda\nu$ to solve for the frequency:

$$\nu = \frac{c}{\lambda}$$

Since c is expressed in meters per second, we must also convert 589 nm to meters.

$$\nu = \left(\frac{2.998 \times 10^{8}\ \cancel{\text{m}}\,\text{s}^{-1}}{589\cancel{\text{nm}}}\right)\left(\frac{1 \times 10^{9}\ \cancel{\text{nm}}}{1\cancel{\text{m}}}\right) = 5.09 \times 10^{14}\,\text{s}^{-1}$$

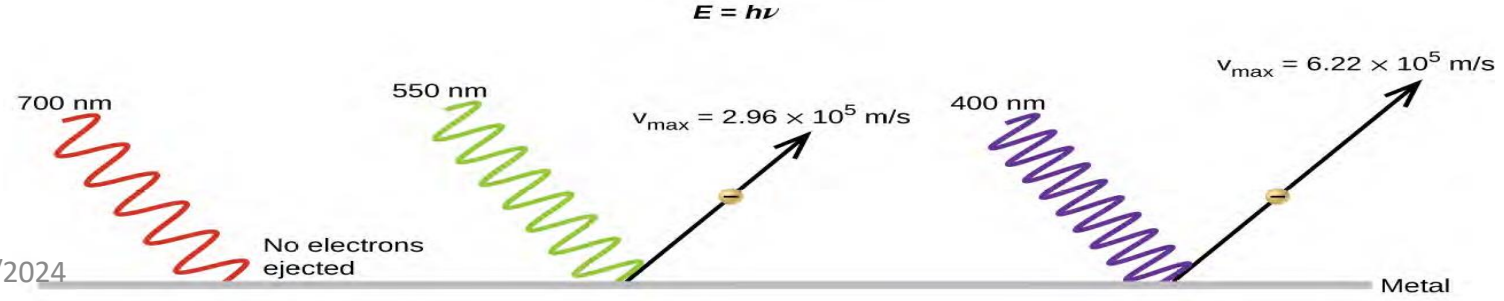
Check Your Learning

One of the frequencies used to transmit and receive cellular telephone signals in the United States is 850 MHz. What is the wavelength in meters of these radio waves?

Answer: 0.353 m = 35.3 cm

5.1.2. Quantization and Photons
The Photoelectric Effect

- Einstein argued that the quantized energies that Planck had postulated in his treatment of blackbody radiation could be applied to the light in the photoelectric effect so that the light striking the metal surface should not be viewed as a wave, but instead as a stream of particles (later called photons) whose energy depended on their frequency.
  - ✓ According to Planck's formula,
    E = hν (or, in terms of wavelength using $c = \nu\lambda$, $E = hc/\lambda$).
  - ✓ Electrons were ejected when hit by photons having sufficient energy (a frequency greater than the threshold).
  - ✓ The greater the frequency, the greater the kinetic energy imparted to the escaping electrons by the collisions.
  - ✓ Light intensity did not depend on the amplitude of the incoming wave, but instead corresponded to the number of photons striking the surface within a given time period.
  - ✓ This explains why the number of ejected electrons increased with increasing brightness, since the greater the number of incoming photons, the greater the likelihood that they would collide with some of the electrons.



Calculating the Energy of Radiation

When we see light from a neon sign, we are observing radiation from excited neon atoms. If this radiation has a wavelength of 640 nm, what is the energy of the photon being emitted?

Solution

We use the part of Planck's equation that includes the wavelength, λ, and convert units of nanometers to meters so that the units of λ and c are the same.

$$E = \frac{hc}{\lambda}$$

$$E = \frac{(6.626 \times 10^{-34}\ \text{J}\cancel{\text{s}})(2.998 \times 10^{8}\ \text{m}\cancel{\text{s}}^{-1})}{(640\cancel{\text{nm}})\left(\frac{1\ \text{m}}{10^{9}\cancel{\text{nm}}}\right)}$$

$$E = 3.10 \times 10^{-19}\ \text{J}$$

Check Your Learning

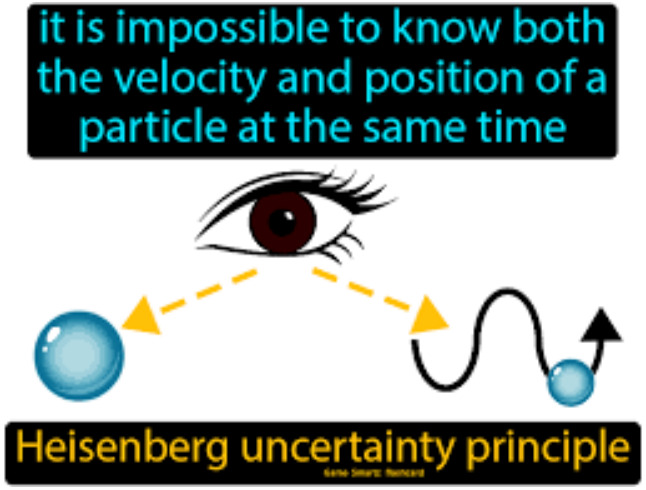
The microwaves in an oven are of a specific frequency that will heat the water molecules contained in food. (This is why most plastics and glass do not become hot in a microwave oven-they do not contain water molecules.) This frequency is about 3×10^{9} Hz. What is the energy of one photon in these microwaves?

Answer: 2×10^{-24} J

5.3. Development of Quantum Theory
5.3.1. The Quantum–Mechanical Model of an Atom

- Proposed by Austrian physicist in 1926 Erwin Schrödinger (1887–1961)
  - ✓ The fundamental idea behind the model is that it's best to abandon the notion of an electron **as a small particle moving around the nucleus in a defined path and to concentrate instead on the electron's wavelike properties.**

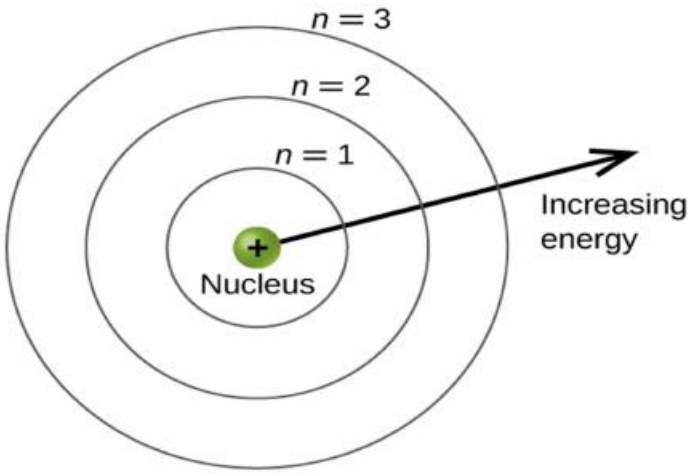
- In fact, it was shown in 1927 by Werner Heisenberg (1901–1976) that it is impossible to know precisely where an electron is and what path it follows—a statement called the Heisenberg uncertainty principle.



5.3.2. Quantum Theory of Electrons in Atoms

A. The principal quantum number (n)

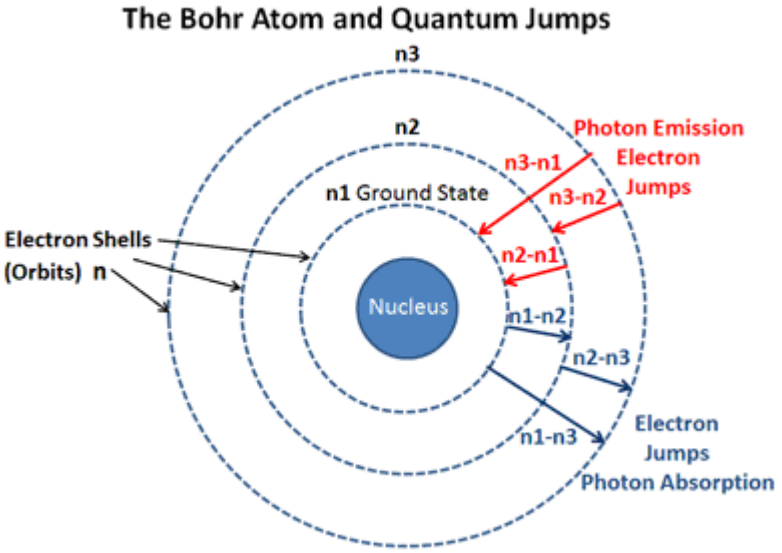
- electrons in atoms **can exist only on discrete energy levels** but not between them
- The energy of **an electron in an atom is quantized**, that is, it can be equal only to certain specific values and
- electrons **can jump from one energy level to another** but not transition smoothly or stay between these levels.
- The energy **levels are labeled** with an n value, where n = 1, 2, 3,….
- electrons that belong to a specific shell are most likely to be found within the corresponding circular area.
- The **further we precede from the nucleus, the higher the shell number, and so the higher the energy leve**l.
- The **protons in the nucleus stabilize the electronic orbitals by electrostatic attraction**
- So the further away the **electron is from the nucleus, the greater the energy** it has.



Electronic transitions,

✓ If the transition is to a higher energy level,
energy is absorbed, and
the energy change has a positive value.

To obtain the amount of energy necessary for the transition to a higher energy level, a photon is absorbed by the atom.



✓ A transition to a lower energy level
involves a release of energy, and
the energy change is negative.

This process is accompanied by emission of a photon by the atom.

$$\Delta E = E final - E initial = -2.18 \times 10^{-1} \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) J$$

B. The angular-momentum quantum number (l)

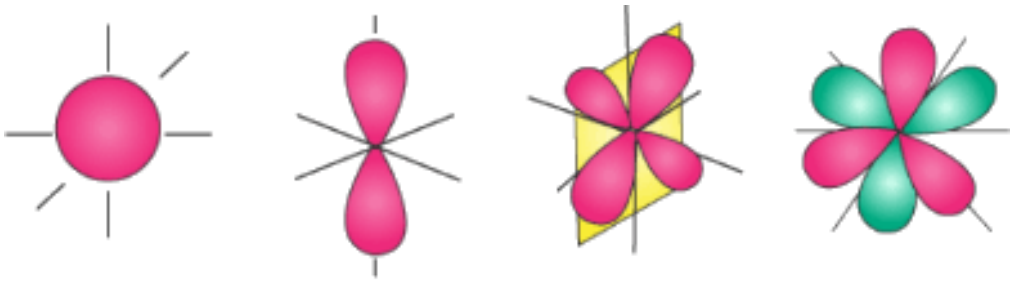
- **the three-dimensional shape of the orbital.**
- For an orbital whose principal quantum number is n, the angular-momentum quantum number l can have any integral value from **0 to n-1**.
- within each shell, **there are n different shapes for orbitals**.

If $n = 1$, then $l = 0$

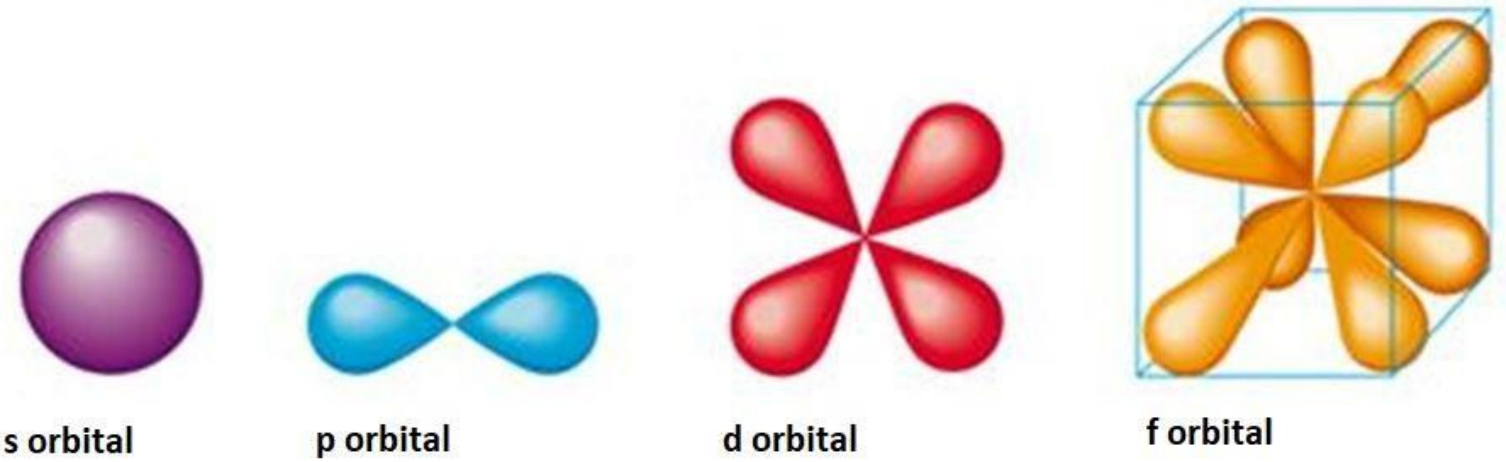
If $n = 2$, then $l = 0$ or 1

If $n = 3$, then $l = 0, 1,$ or 2

- Orbitals with **l = 0 are called s orbitals and they make up the s subshells**.
- The **value l = 1 corresponds to the p orbitals**. For a given n, p orbitals constitute a p subshell (e.g., **3p if n = 3**).
- The orbitals **with l = 2 are called the d orbitals**, followed by the f-,g-, and h-orbitals for l = 3, 4, and 5.



Shapes of s, p, d, and f orbitals



C. The magnetic quantum number m_l

- the **spatial orientation of the orbital with respect to a standard set of coordinate axes**.
- For an orbital whose angular-momentum quantum number is l, the magnetic quantum number **m_l can have any integral value from -l to +l**
- there are 2l + 1 different spatial orientations for those orbitals.

If $l = 0$, then $m_l = 0$

If $l = 1$, then $m_l = -1, 0,$ or $+1$

If $l = 2$, then $m_l = -2, -1, 0, +1,$ or $+2$

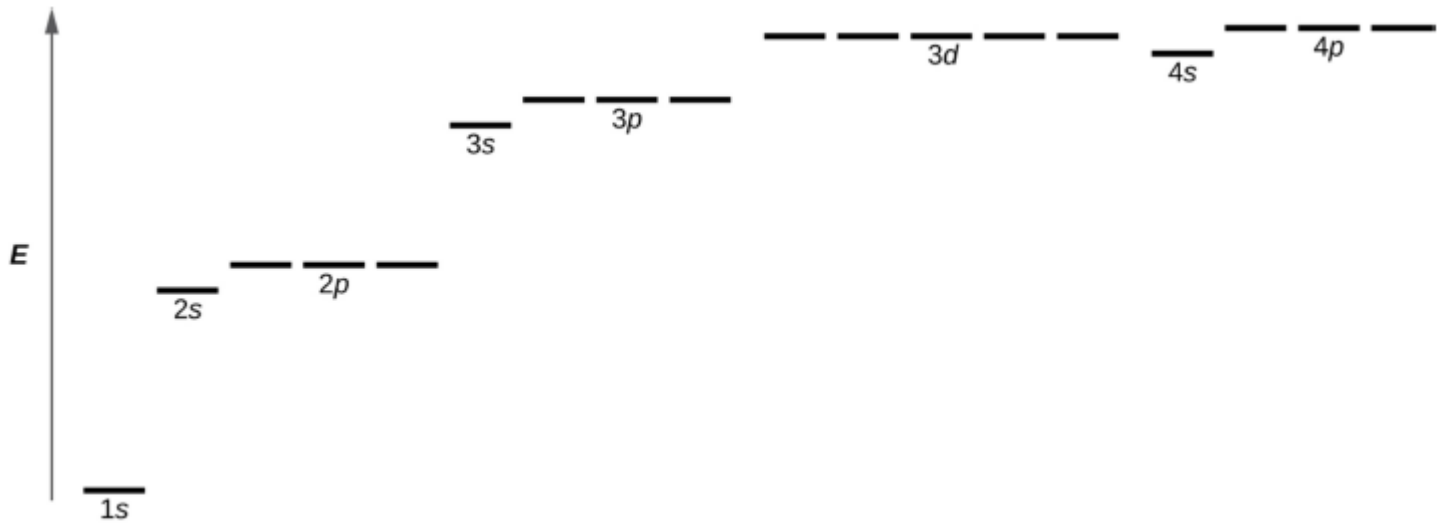
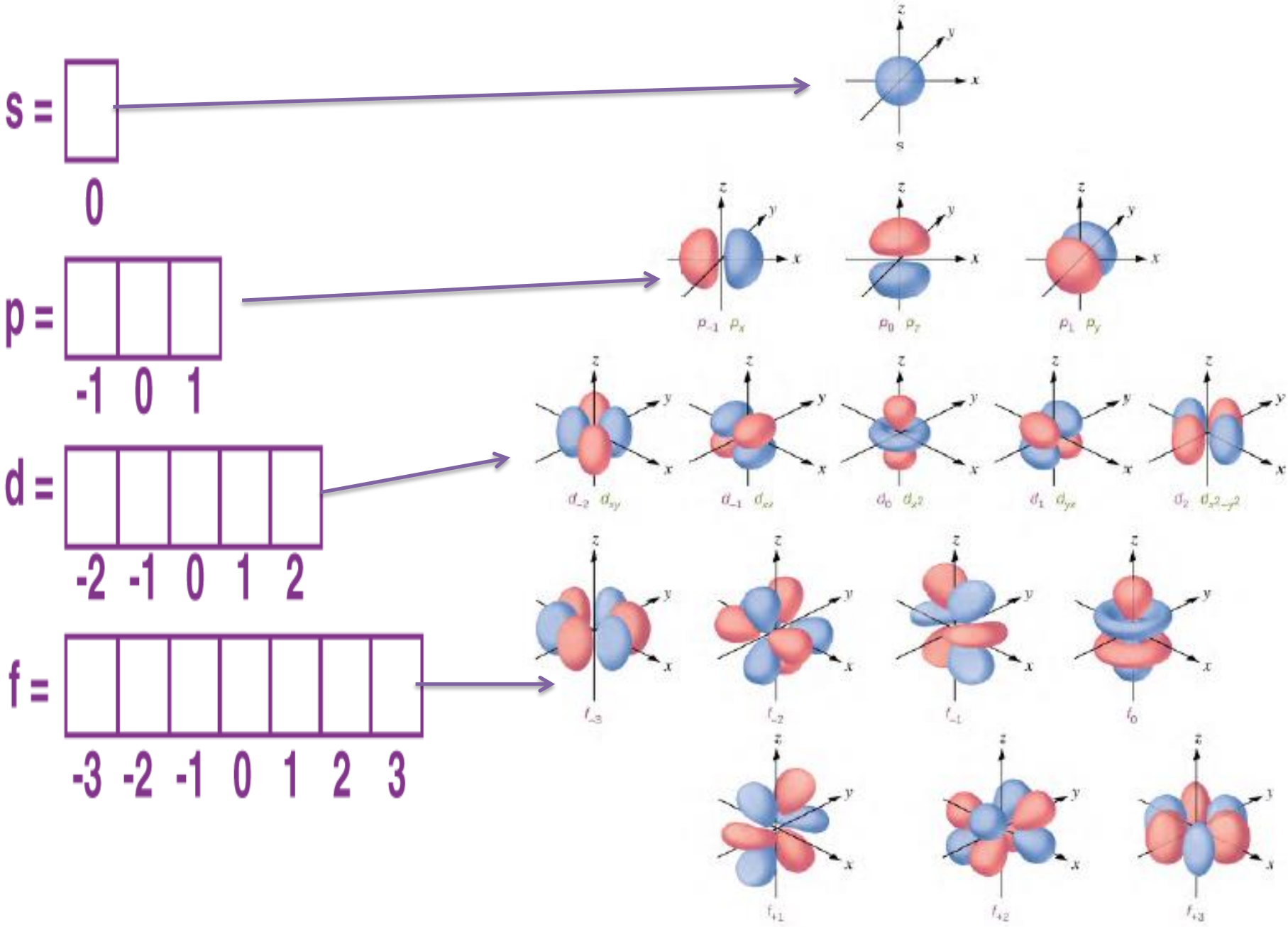


Figure 5.9: The chart shows the energies of electron orbitals in a multi-electron atom.

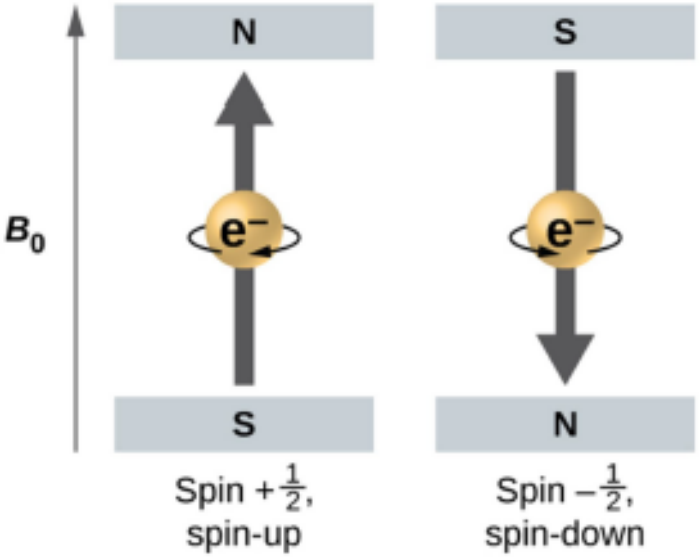

s =
0
p =
-1 0 1
d =
-2 -1 0 1 2
f =
-3 -2 -1 0 1 2 3

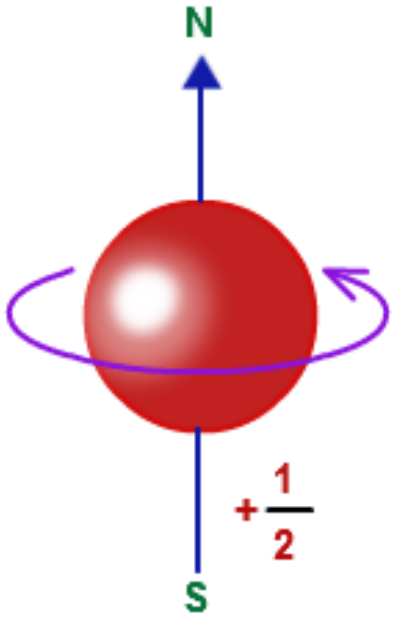
n	l	m_l	Orbital Notation	Number of Orbitals in Subshell	Number of Orbitals in Shell
1	0	0	1*s*	1	1
2	0	0	2*s*	1	4
	1	−1, 0, +1	2*p*	3	
3	0	0	3*s*	1	9
	1	−1, 0, +1	3*p*	3	
	2	−2, −1, 0, +1, +2	3*d*	5	
4	0	0	4*s*	1	16
	1	−1, 0, +1	4*p*	3	
	2	−2, −1, 0, +1, +2	4*d*	5	
	3	−3, −2, −1, 0, +1, +2, +3	4*f*	7	

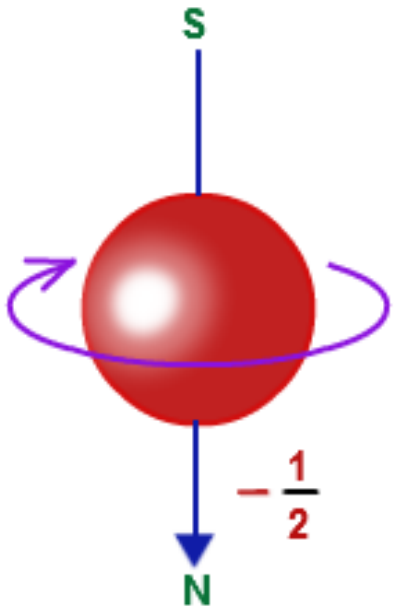
The three quantum numbers, **n, l, and m_l**, (define the energy, shape, and spatial orientation of orbitals) are properties of specific atomic orbitals that also define in what part of the space an electron is most likely to be located. Orbitals are a result of solving the Schrödinger equation for electrons in atoms.

D. Spin quantum number m_s

- electrons are **free to spin in either a clockwise or a counterclockwise direction**.
- This **spinning charge gives rise to a tiny magnetic field**
- can have either of two values **+1/2 (↑) or -1/2 (↓)**







5.3.3. The Pauli Exclusion Principle

- An electron in an atom is completely **described by four quantum numbers: n, l, m_l, and m_s.**
- formulated as follows: **No two electrons in the same atom can have exactly the same set of all the four quantum numbers**.
- Electrons that occupy the same orbital have **the same three quantum numbers, n, l, and m_l**
- an orbital **can hold only two electrons**, which must have opposite spins

Quantum Numbers, Their Properties, and Significance

Name	Symbol	Allowed values	Physical meaning
principle quantum number	n	1, 2, 3, 4, ...	shell, the general region for the value of energy for an electron on the orbital
angular momentum or azimuthal quantum number	l	$0 \le l \le n-1$	subshell, the shape of the orbital
magnetic quantum number	m_l	$-l \le m_l \le l$	orientation of the orbital
spin quantum number	m_s	$\frac{1}{2}, -\frac{1}{2}$	direction of the intrinsic quantum "spinning" of the electron

Working with Shells and Subshells

Indicate the number of subshells, the number of orbitals in each subshell, and the values of l and m_l for the orbitals in the $n = 4$ shell of an atom.

Solution

For $n = 4$, l can have values of 0, 1, 2, and 3. Thus, *s*, *p*, *d*, and *f* subshells are found in the $n = 4$ shell of an atom. For $l = 0$ (the *s* subshell), m_l can only be 0. Thus, there is only one 4*s* orbital. For $l = 1$ (*p*-type orbitals), m can have values of –1, 0, +1, so we find three 4*p* orbitals. For $l = 2$ (*d*-type orbitals), m_l can have values of –2, –1, 0, +1, +2, so we have five 4*d* orbitals. When $l = 3$ (*f*-type orbitals), m_l can have values of –3, –2, –1, 0, +1, +2, +3, and we can have seven 4*f* orbitals. Thus, we find a total of 16 orbitals in the $n = 4$ shell of an atom.

Check Your Learning

Identify the subshell in which electrons with the following quantum numbers are found: (a) $n = 3$, $l = 1$; (b) $n = 5$, $l = 3$; (c) $n = 2$, $l = 0$.

Answer: (a) 3*p* (b) 5*f* (c) 2*s*

Maximum Number of Electrons

Calculate the maximum number of electrons that can occupy a shell with (a) $n = 2$, (b) $n = 5$, and (c) n as a variable. Note you are only looking at the orbitals with the specified n value, not those at lower energies.

Solution

(a) When $n = 2$, there are four orbitals (a single 2*s* orbital, and three orbitals labeled 2*p*). These four orbitals can contain eight electrons.

(b) When $n = 5$, there are five subshells of orbitals that we need to sum:

1 orbitals labeled 5*s*
3 orbitals labeled 5*p*
5 orbitals labeled 5*d*
7 orbitals labeled 5*f*

+9 orbitals labeled 5*g*
25 orbitals total

Again, each orbital holds two electrons, so 50 electrons can fit in this shell.

(c) The number of orbitals in any shell *n* will equal n^2. There can be up to two electrons in each orbital, so the maximum number of electrons will be $2 \times n^2$

Check Your Learning

If a shell contains a maximum of 32 electrons, what is the principal quantum number, *n*?

Answer: $n = 4$

Working with Quantum Numbers

Complete the following table for atomic orbitals:

Orbital	n	l	m_l degeneracy	Radial nodes (no.)
4*f*				
	4	1		
	7		7	3
5*d*				

Radial nodes = n – l - 1

Solution

The table can be completed using the following rules:

- The orbital designation is nl, where $l = 0, 1, 2, 3, 4, 5, \ldots$ is mapped to the letter sequence s, *p*, *d*, *f*, *g*, *h*, …,
- The m_l degeneracy is the number of orbitals within an l subshell, and so is $2l + 1$ (there is one *s* orbital, three *p* orbitals, five *d* orbitals, seven *f* orbitals, and so forth).
- The number of radial nodes is equal to $n - l - 1$.

Orbital	*n*	*l*	m_l degeneracy	Radial nodes (no.)
4*f*	4	3	7	0
4*p*	4	1	3	2
7*f*	7	3	7	3
5*d*	5	2	5	2

Check Your Learning

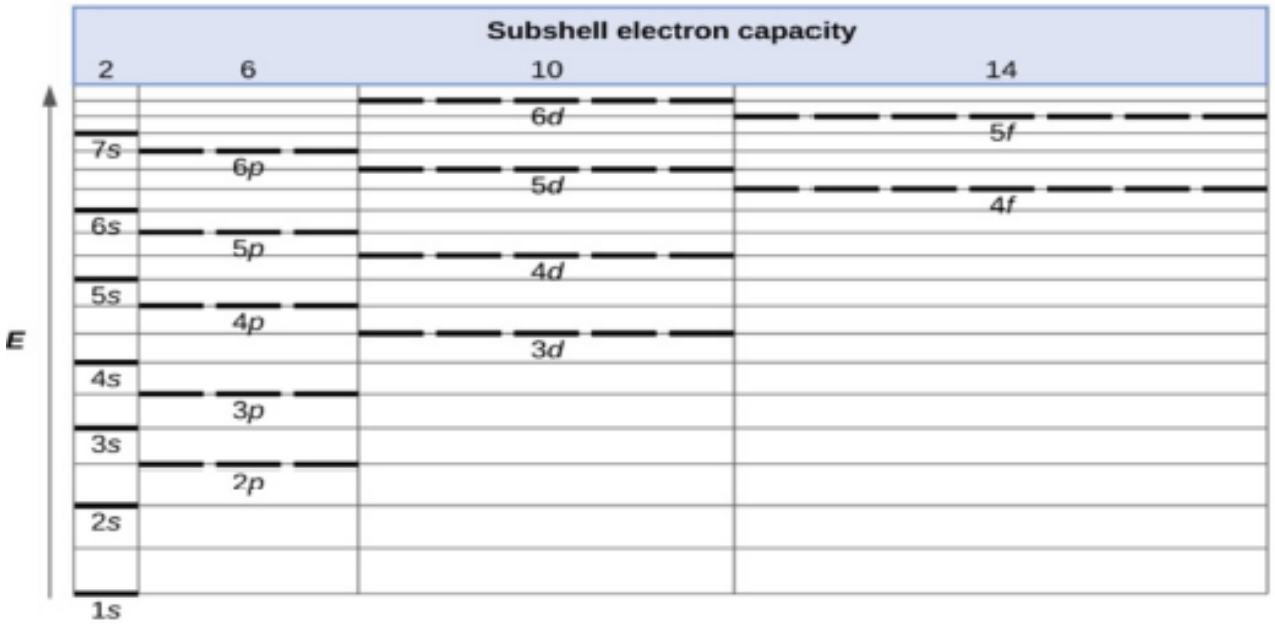
How many orbitals have $l = 2$ and $n = 3$?

Answer: The five degenerate 3*d* orbitals

5.4 Electronic Structure of Atoms (Electron Configurations)

5.4.1. Orbital Energies and Atomic Structure

- The energy of atomic orbitals increases as the principal quantum number, n, increases.
- the energy of the orbitals increases within a shell in the order s<p<d<f.
- The 1s orbital is the orbital with electrons of lowest energy.
- The energy increases as we move up to the 2s and then 2p, 3s, and 3p orbitals
- the increasing n value has more influence on energy than the increasing l value for small atoms.
- For larger atoms, the 3d orbital is higher in energy than the 4s orbital.



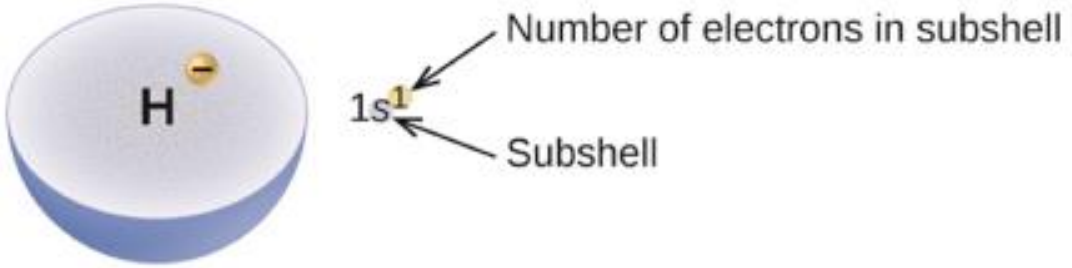
Electron configuration: the arrangement of electrons in the orbitals of an atom.

- Described by a symbol that contains three pieces of information :

1. The number of the principal quantum shell, n,
2. The letter that designates the orbital type (the subshell, l), and
3. A superscript number that designates the number of electrons in that particular subshell.

Example 1, the notation $2p^4$ (read "two–p–four")
indicates four electrons in a p subshell (l = 1)
with a principal quantum number (n) of 2.

Example 2: the notation $3d^8$ (read "three–d–eight")
indicates eight electrons in the d subshell (i.e., l = 2)
with a principal shell for which n = 3.



5.4.2. The Aufbau Principle

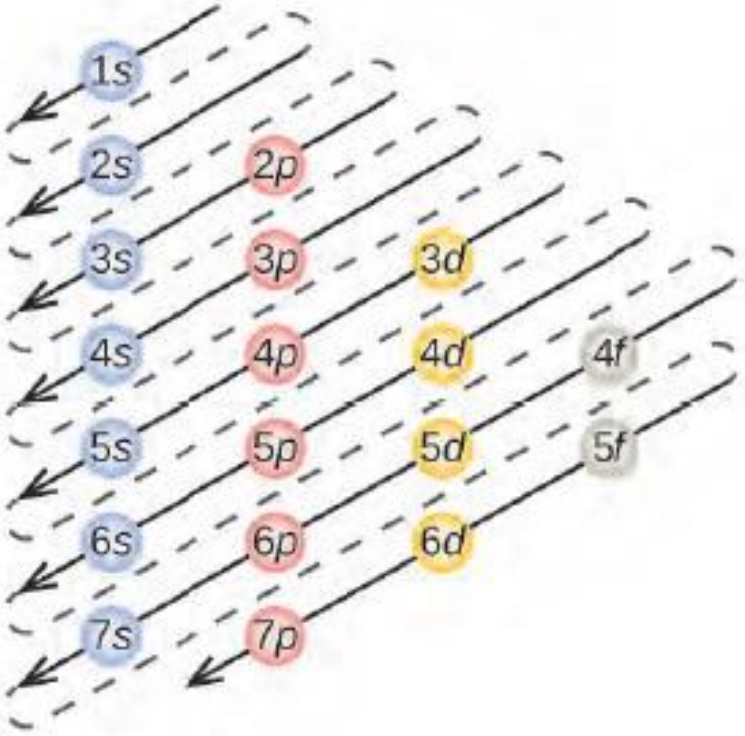
- Guides the filling order of orbitals

1. Lower-energy orbitals fill before higher-energy orbitals.
2. An orbital can hold only two electrons, which must have opposite spins. This is just a restatement of the **Pauli exclusion principle,** emphasizing that no two electrons in an atom can have the same four quantum numbers.
3. If two or more degenerate orbitals are available, one electron goes into each until all are half-full, a statement called **Hund's rule**

Hydrogen: Hydrogen has only one electron, which must go into the lowest energy, 1*s* orbital.

H $1s^1$ 

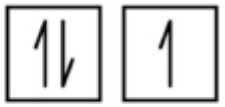
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Helium: Helium has two el energy, 1*s* orbital. The two electrons have opposit

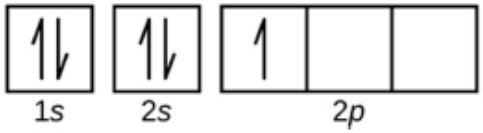
He $1s^2$ 

Lithium and beryllium: Wi urth electrons go into the next available, 2*s* orbital.

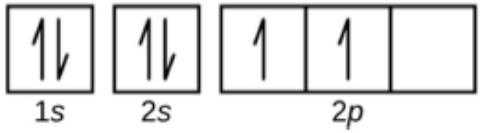
Li $1s^2 2s^1$ 

***Boron through neon**:* have their three 2*p* orbitals filled successively. Because these three 2*p* orbitals have the same energy, they are **degenerate** and are filled according to Hund's rule.

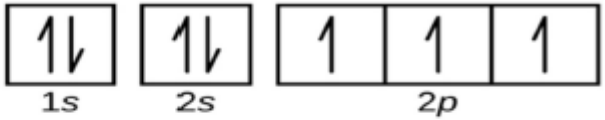
B $1s^2 2s^2 2p^1$



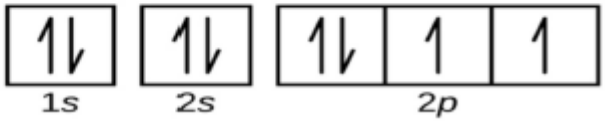
C $1s^2 2s^2 2p^2$



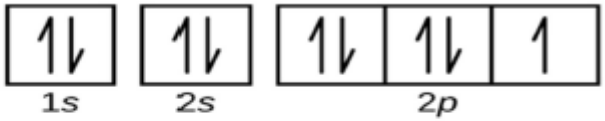
N $1s^2 2s^2 2p^3$



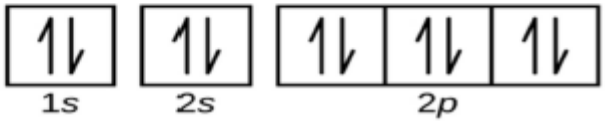
O $1s^2 2s^2 2p^4$



F $1s^2 2s^2 2p^5$



Ne $1s^2 2s^2 2p^6$



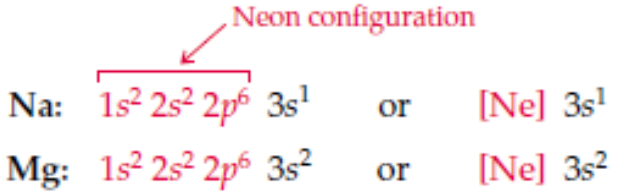
Note that we often
write the configurations in a shorthand version by giving the symbol of the noble gas in the previous row to indicate electrons in filled shells and then specifying only those electrons in unfilled shells.

- The electrons occupying the outermost shell orbital(s) (highest value of n) are called **valence electrons**, and those occupying the inner shell orbitals are called **core electrons.**
- core electron shells correspond to noble gas electron configurations,
- core electron configuration can be replaced by electron configurations of the noble gas.

Example: For our sodium example, the symbol [Ne] represents core electrons, ($1s^2 2s^2 2p^6$) and the abbreviated or condensed configuration is [Ne]$3s^1$.



Sodium and magnesium: The 3*s* orbital is filled next, giving sodium and magnesium the ground-state electron configurations



Aluminum through argon: The 3*p* orbitals are now filled according to the same rules used previously for filling the 2*p* orbitals of boron through neon.

Al: [Ne] $3s^2\,3p^1$ Si: [Ne] $3s^2\,3p^2$ P: [Ne] $3s^2\,3p^3$

S: [Ne] $3s^2\,3p^4$ Cl: [Ne] $3s^2\,3p^5$ Ar: [Ne] $3s^2\,3p^6$

Elements past argon: the next two electrons in potassium and calcium go into the 4*s* subshell. Only then does filling of the 3*d* subshell occur to give the first transition metal series from scandium through zinc.

K: [Ar] $4s^1$ Ca: [Ar] $4s^2$ Sc: [Ar] $4s^2\,3d^1 \longrightarrow$ Zn: [Ar] $4s^2\,3d^{10}$

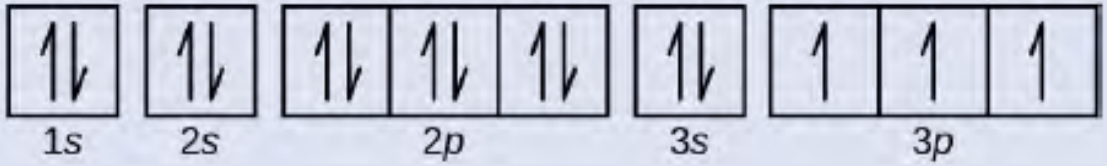
Quantum Numbers and Electron Configurations

What is the electron configuration and orbital diagram for a phosphorus atom? What are the four quantum numbers for the last electron added?

Solution

The atomic number of phosphorus is 15. Thus, a phosphorus atom contains 15 electrons. The order of filling of the energy levels is 1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, . . . The 15 electrons of the phosphorus atom will fill up to the 3*p* orbital, which will contain three electrons:

P $1s^2 2s^2 2p^6 3s^2 3p^3$



The last electron added is a 3*p* electron. Therefore, $n = 3$ and, for a *p*-type orbital, $l = 1$. The m_l value could be –1, 0, or +1. The three *p* orbitals are degenerate, so any of these m_l values is correct. For unpaired electrons, convention assigns the value of $+\frac{1}{2}$ for the spin quantum number; thus, $m_s = +\frac{1}{2}$.

Check Your Learning

Identify the atoms from the electron configurations given:

(a) $[Ar]4s^2 3d^5$

(b) $[Kr]5s^2 4d^{10} 5p^6$

Answer: (a) Mn (b) Xe

5.4.3. Electron Configurations and the Periodic Table

- Elements in any one group (or column) have the same number of valence electrons;
  - ✓ the alkali metals lithium and sodium each have only one valence electron,
  - ✓ the alkaline earth metals beryllium and magnesium each have two,
  - ✓ the halogens fluorine and chlorine each have seven valence electrons.
- The similarity in chemical properties among elements of the same group occurs because they have the same number of valence electrons.
- It is the loss, gain, or sharing of valence electrons that defines how elements react.

❖ **Main group elements** (sometimes called representative elements) are those in which the last electron added enters an s or a p orbital in the outermost shell.

- ✓ includes all the nonmetallic elements, as well as many metals and the metalloids

❖ **Transition elements or transition metals**: are metallic elements in which the last electron added enters a d orbital.

- ✓ The valence electrons (those added after the last noble gas configuration) in these elements include the ns and (n – 1) d electrons.
- ✓ Also called as d-block elements

❖ **Inner transition elements** are metallic elements in which the last electron added occupies an f orbital. Their valance shells of the (n – 2)f, the (n – 1)d, and the ns subshells.

- ✓ There are two inner transition series:
  - a. The lanthanide series: lanthanide (La) through lutetium (Lu)
  - b. The actinide series: actinide (Ac) through lawrencium (Lr)

5.4.4. Electron Configurations of Ions

- Ions are formed when atoms gain or lose electrons.
- **A cation** forms when electrons are removed from a parent atom.
  - ✓ **For main group elements**, the electrons that were added last are the first electrons removed.
  - ✓ **For transition metals and inner transition metals**, electrons in the s orbital are easier to remove than the d or f electrons, and so the highest ns electrons are lost, and then the (n – 1)d or (n – 2)f electrons are removed.
- **An anion** forms when one or more electrons are added to a parent atom.
- The added electrons fill in the order predicted by the Aufbau principle.

Predicting Electron Configurations of Ions

What is the electron configuration and orbital diagram of:

(a) Na^{+}

(b) P^{3-}

(c) Al^{2+}

(d) Fe^{2+}

(e) Sm^{3+}

Solution

(a) Na: $1s^2 2s^2 2p^6 3s^1$. Sodium cation loses one electron, so Na^+: $1s^2 2s^2 2p^6 3s^1 = Na^+$: $1s^2 2s^2 2p^6$.

(b) P: $1s^2 2s^2 2p^6 3s^2 3p^3$. Phosphorus trianion gains three electrons, so P^{3-}: $1s^2 2s^2 2p^6 3s^2 3p^6$.

(c) Al: $1s^2 2s^2 2p^6 3s^2 3p^1$. Aluminum dication loses two electrons Al^{2+}: $1s^2 2s^2 2p^6 3s^2 3p^1 =$ Al^{2+}: $1s^2 2s^2 2p^6 3s^1$.

(d) Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$. Iron(II) loses two electrons and, since it is a transition metal, they are removed from the 4*s* orbital Fe^{2+}: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6 = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$.

(e). Sm: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^6$. Samarium trication loses three electrons. The first two will be lost from the 6*s* orbital, and the final one is removed from the 4*f* orbital. Sm^{3+}: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^6 = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 4f^5$.

Check Your Learning

Which ion with a +2 charge has the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^5$? Which ion with a +3 charge has this configuration?

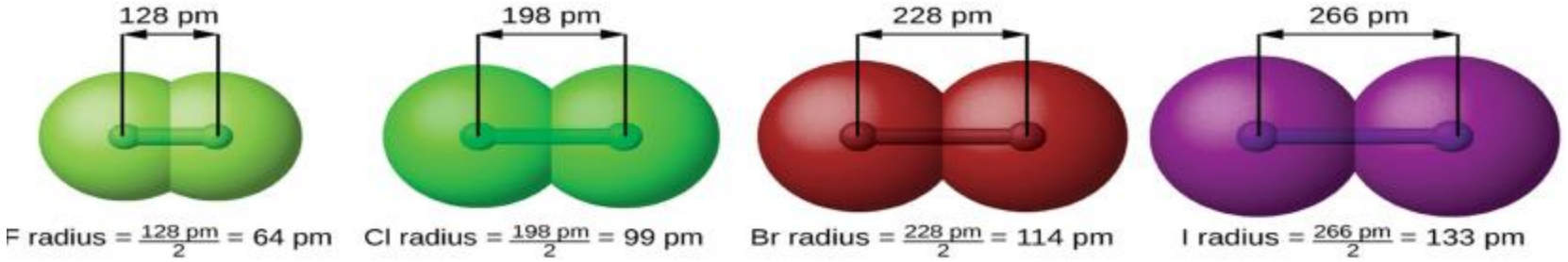
Answer: Tc^{2+}, Ru^{3+}

5.5. Periodic Variation in Element Properties

5.5.1. Variation in Covalent Radius

- **Covalent Radii (Atomic radii) *increase* going down a group of the periodic table**
  - ✓ Because successively larger valence-shell orbitals are occupied.

Atom	Covalent radius (pm)	Nuclear charge
F	64	+9
Cl	99	+17
Br	114	+35
I	133	+53
At	148	+85



- **Atomic radii *decrease* going across a row from left to right**
  - ✓ Because of an increase in effective nuclear charge (Z_{eff}) for the valence-shell electrons.
  - ✓ The effective nuclear charge actually felt by an electron, is lower than the true nuclear charge *Z* because of shielding by other electrons in the atom.
  - ✓ Thus, each time we move from one element to the next across a period, Z increases by one, but the shielding increases only slightly. Thus, Z_{eff} increases as we move from left to right across a period.
  - ✓ The stronger pull (higher effective nuclear charge) experienced by electrons on the right side of the periodic table draws them closer to the nucleus, making the covalent radii smaller

$$Z_{eff} = Z - shielding$$

Sorting Atomic Radii

Predict the order of increasing covalent radius for Ge, Fl, Br, Kr.

Solution

Radius increases as we move down a group, so Ge < Fl (Note: Fl is the symbol for flerovium, element 114, NOT fluorine). Radius decreases as we move across a period, so Kr < Br < Ge. Putting the trends together, we obtain Kr < Br < Ge < Fl.

5.5.2. Variation in Ionic Radii

- A cation always has fewer electrons and the same number of protons as the parent atom;
  - ✓ it is smaller than the atom from which it is derived.
  - ✓ shells experience a greater effective nuclear charge Z_{eff}

  For example, the covalent radius of an aluminum atom ($1s^2 2s^2 2p^6 3s^2 3p^1$) is 118 pm, whereas the ionic radius of an Al^{3+} ($1s^2 2s^2 2p^6$) is 68 pm.
- Cations with larger charges are smaller than cations with smaller charges

  Example: V^{2+} has an ionic radius of 79 pm, while that of V^{3+} is 64 pm

- An anion is formed by the addition of one or more electrons to the valence shell of an atom.
- results in a greater repulsion among the electrons and a decrease in Z_{eff} per electron.
- Because of the increased number of electrons and the decreased Z_{eff} the radius of an anion is larger than that of the parent atom

  For example, a sulfur atom ($[Ne]3s^2 3p^4$) has a covalent radius of 104 pm, whereas the ionic radius of the sulfide anion ($[Ne]3s^2 3p^6$) is 170 pm.



Isoelectronic : Atoms and ions that have the same electron configuration

Examples: N^{3-}, O^{2-}, F^-, Ne, Na^+, Mg^{2+}, and Al^{3+} ($1s^2 2s^2 2p^6$).

P^{3-}, S^{2-}, Cl^-, Ar, K^+, Ca^{2+}, and Sc^{3+} ($[Ne]3s^2 3p^6$)

5.5.3. Variation in Ionization Energies

First ionization energy (IE_1): is the amount of energy required to remove the most loosely bound electron from a gaseous atom in its ground state.

$$X(g) \longrightarrow X+(g) + e^- \quad IE1$$

Second ionization energy (IE_2): is the energy required to remove the second most loosely bound electron.

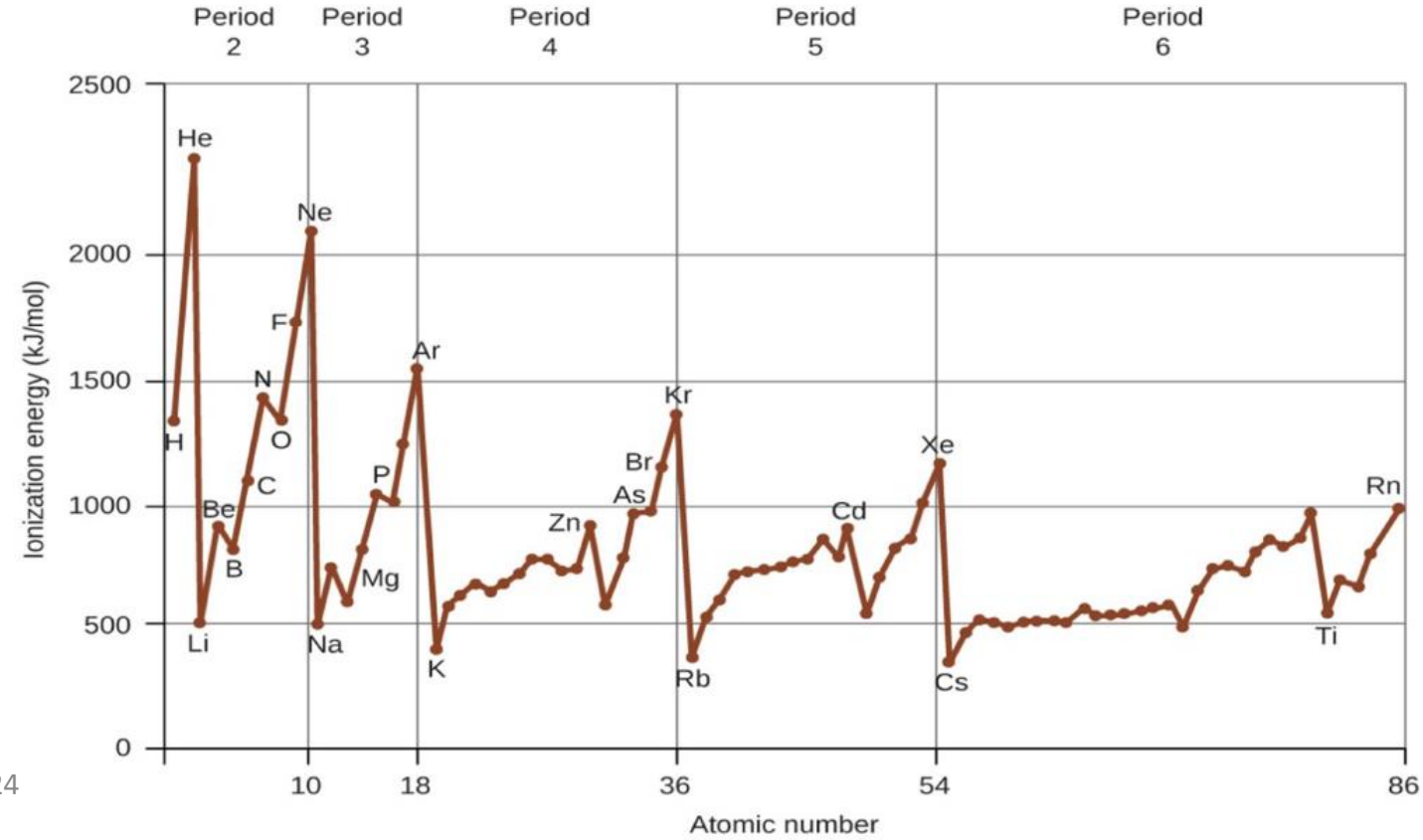
$$X+(g) \longrightarrow X_2+(g) + e^- \quad IE_2$$

- Ionization processes are **endothermic** and
- IE values are **always positive**.
- For **larger atoms**, the most loosely bound electron easily removed. Since it is located farther from the nucleus and.
  - ✓ Thus, as **size (atomic radius) increases**, the ionization energy should decrease.
  - ✓ The first ionization energies are **decreased down a group** and **increased across a period**.

- Within any one shell, an **s electron is harder** to remove from an atom than a **p electron** in the same shell.

❖ Deviation from the trends

Example 1: The electron removed during the ionization of beryllium ($[He]2s^2$) is an s electron, whereas the electron removed during the ionization of boron ($[He]2s^2 2p^1$) is a p electron; this results in a lower first ionization energy for boron, even though its nuclear charge is greater by one proton.



Example 2: The first ionization energy for oxygen is slightly less than that for nitrogen, despite the trend in increasing IE_1 values across a period.

Removing one electron from oxygen will eliminate the electron–electron repulsion caused by pairing the electrons in the 2p orbital and will result in a half-filled orbital (which is energetically favorable).

O $1s^2 2s^2 2p^4$



- Removing an electron from a cation is more difficult than removing an electron from a neutral atom because of the greater electrostatic attraction to the cation.
- Removing an electron from a cation with a higher positive charge is more difficult than removing an electron from an ion with a lower charge.

Element	IE_1	IE_2	IE_3	IE_4	IE_5	IE_6	IE_7
K	418.8	3051.8	4419.6	5876.9	7975.5	9590.6	11343
Ca	589.8	1145.4	4912.4	6490.6	8153.0	10495.7	12272.9
Sc	633.1	1235.0	2388.7	7090.6	8842.9	10679.0	13315.0
Ga	578.8	1979.4	2964.6	6180	8298.7	10873.9	13594.8
Ge	762.2	1537.5	3302.1	4410.6	9021.4	Not available	Not available
As	944.5	1793.6	2735.5	4836.8	6042.9	12311.5	Not available

5.5.4. Variation in Electron Affinities

The electron affinity (EA) is the energy change for the process of adding an electron to a gaseous atom to form an anion (negative ion).

$$X(g) + e^- \longrightarrow X^-(g) \qquad EA_1$$

EA can be either endothermic or exothermic, depending on the element.

- Adding an electron across a series of atoms is easier, as the effective nuclear charge of the atoms increases. Therefore, we go from left to right across a period, EAs tend to become more negative.
- The exceptions found among the elements of group 2A, group 5A, and group 8A can be understood based on the electronic structure of these groups.
  - ✓ Group 8A, have a completely filled shell and the incoming electron must be added to a higher n level, which is more difficult to do.
  - ✓ Group 2A has a filled ns subshell, and so the next electron added goes into the higher energy np.
  - ✓ Group 5A has a half-filled np subshell and the next electron must be paired with an existing np electron.

➢ As we move down a group, we see that the second element in the group most often has the most negative EA.

✓ This can be attributed to the small size of the n = 2 shell and the resulting large electron – electron repulsions.

✓ **For example**, chlorine, with an EA value of –348 kJ/mol, has the highest value of any element in the periodic table.

✓ The EA of fluorine is –322 kJ/mol. When we add an electron to a fluorine atom to form a fluoride anion (F^-), we add an electron to the n = 2 shell.

✓ The electron is attracted to the nucleus, but there is also significant repulsion from the other electrons already present in this small valence shell.

✓ The chlorine atom has the same electron configuration in the valence shell, but because the entering electron is going into the n = 3 shell, it occupies a considerably larger region of space and the electron–electron repulsions are reduced.

✓ The entering electron does not experience as much repulsion and the chlorine atom accepts an additional electron more readily, resulting in a more negative EA.

Electron Affinity Values for Selected Elements (kJ/mol)

Period	Group 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H –72																	He +20*
2	Li –60	Be +240*											B –23	C –123	N 0	O –141	F –322	Ne –30
3	Na –53	Mg +230*											Al –44	Si –120	P –74	S –200	Cl –348	Ar +35*
4	K –48	Ca +150*	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga –40*	Ge –115	As –7	Se –195	Br –324	Kr +40*
5	Rb –46	Sr +160*	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In –40*	Sn –121	Sb –101	Te –190	I –295	Xe +40*
6	Cs –45	Ba +50*	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl –50	Pb –101	Bi –101	Po –170	At –270*	Rn +40*
7	Fr	Ra																

* Calculated value